1) What is the purpose of a loop structure?
   A loop repeats a block of code while a statement is true, or for a specific number of times.

2) Explain the difference between a while statement and a do-while statement.
   While checks the condition before executing the loop, so if the condition isn't there, it won't execute at all

   Do-while executes the loop, then checks the condition, so it will always execute at least once.

3) An input validation loop is a loop that checks user input for valid data. If valid data is not entered, the loop iterates until valid data is entered. In which review of this chapter did you write code for input validation loops?
   I wrote input validation loops in both mastery projects that I completed for chapter 5. I used them for error handling, and to ensure the code wouldn't break down.

4)
   a) What is an infinite loop?
      A loop that continues forever
   b) List two types of errors that can lead to an infinite loop.
      Incorrect loop condition or failure to update loop control variable.
   c) What is meant by overflow?
      Overflow is when a value exceeds the maximum or minimum value that its data type can store.

5) How many times will the do-while loop execute?

```
int x = 0;
do {
        x = x + 2;
} while (x < 120);
```

   60 times

6) What initial value of x would make the loop infinite?

```
do {
        x = x - 3;
} while (x < 120);
```

   Any initial value less than 123

7) Compare and contrast counters and accumulators. List two uses for each.
   Counter increases or decreases by a set amount
   - Counting the number of students

- Counting the number of loop iterations

Accumulator stores a running total by adding values that may vary
- Calculating the sum of numbers
- Calculating the total cost of items

8) Write a for statement that sums the integers from 3 to 10, inclusive

```
int sum = 0;
int i = 3;

for (; i <= 10; i++)
{
        sum += i;
}
```

9) List two factors that should be considered when determining which loop structure to choose.
- Whether the number of repetitions is known
- Whether the loop must execute at least once.

11) Consider the following assignment:
String x = “my string”;

Determine the value returned by each of the following methods:
a) x.length() 9
b) x.substring(0, 3) “ my “
c) x.toLowerCase() “my string”
d) x.toUpperCase “MY STRING”
e) x.trim() “my string”